

Sigma 6.0 (DSA)

90
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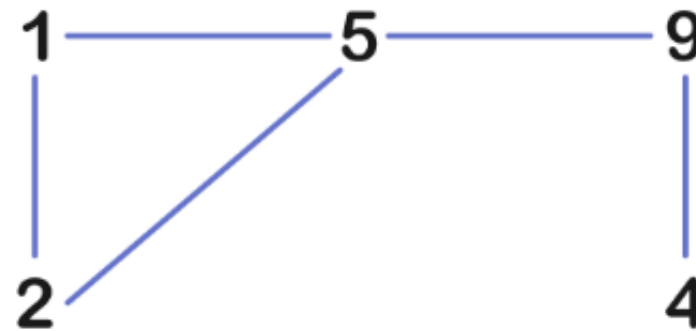
1 of 5

Automatic Zoom

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Articulation Point

An articulation point (or cut vertex) is a vertex in an undirected connected graph if removing it (and edges through it) disconnects the graph. Articulation points are single points in a connected network that if they fail, split the network into two or more components. They are useful in the design of dependable networks. An articulation point in a disconnected undirected graph is a vertex removal that increases the number of connected components..



In this 5 and 9 are articulation points.



1 and 2 Articulation Points

Articulation Points in a Graph using Tarjan's Algorithm:

The idea is to use DFS (Depth First Search). In DFS, follow vertices in a tree form called the DFS tree. In the DFS tree, a vertex u is the parent of another vertex v , if v is discovered by u .

In DFS tree, a vertex u is an articulation point if one of the following two conditions is true.

- u is the root of the DFS tree and it has at least two children.
- u is not the root of the DFS tree and it has a child v such that no vertex in the subtree